Chapter 1 - Introduction (early draft)

Drafted ahead of the E13 sequence, which places the Introduction last. Provisional; to be revised after research-question sign-off.

Status of this chapter. This draft is written *ahead of its place in the writing order*: the confirmed chapter structure places the Introduction last, once the research questions are signed off. It is therefore provisional in a stronger sense than the rest of the proposal — it summarises questions whose wording is pending decisions D-RQ-01 and D-RQ-02, and it will be rewritten rather than edited after sign-off. Two caveats carry through: references to the 2026-08-05 supervision discussion derive from a machine transcript with inferred, undiarised speakers, pending the participants' confirmation; and the planned literature searches (QL-01–QL-05) are specified but not executed, so nothing here about the state of the literature is a finding.

1.1 Context and motivation

Conceptual modelling asks a person to turn a natural-language description of a domain into a structured artefact: a class diagram, a use-case diagram, a process specification. The task looks determinate and is not. A description under-determines its model, and two competent modellers reading the same text will disagree about whether a concept becomes a class or an attribute, whether an association carries a role name, whether a secondary actor belongs in the diagram at all — and both can be defensibly right.

The same holds wherever a written norm must be applied to a particular case, an activity this proposal calls **guideline operationalization**: turning a general guideline into concrete decisions about a specific artefact. Modelling guidelines in a course, documentation standards in an engineering organisation, and care guidelines in a regulated clinical process share the structural feature that legitimate application admits more than one correct answer.

At scale — an instructor assessing well over a hundred student models, an organisation reviewing design documents against a standard — this forces an unavoidable demand: separate variation that is *meaningful*, a systematic and defensible alternative reading of the guideline, from variation that is error or noise. Errors run both ways. Penalising legitimate alternatives teaches people that the standard is arbitrary; accepting genuine errors empties the guideline of force.

The VEGO-AI framework (Reinhartz-Berger, Bragilovski & Sturm, MODELS '26, DOI 10.1145/3822455.3830312) makes this concrete and is the motivating case for this research. Four cooperating LLM agents — a Language Advisor, a Domain Advisor, a Model Inspector, and a Variability Explorer — distinguish **substantial** variability (systematic, meaningful, legitimate alternatives, which should feed back into the guideline) from **occasional** variability (sporadic or erroneous deviation). It was evaluated on 178 student case models across two domains and two modelling languages, using GPT-4o, and gives this doctorate an instrumented setting in which the interpretive problem is already isolated.

1.2 The problem: detection is not interpretation

Automated assessment has become competent at detection: comparing an artefact against a guideline and reporting where they differ is broadly a tractable engineering task. What is not tractable is the step immediately after — deciding what a detected deviation *means*.

A recurring difference between a student's model and the reference may be a legitimate alternative design; a plain error; a domain convention the guideline never wrote down; a defect in the guideline itself; or a genuine ambiguity on which qualified experts would split. These readings are not separable from the surface form of the deviation. Telling them apart requires knowledge that is not in the artefact at all — how the guideline is meant to be read, what the domain permits, what the institution has previously accepted. That knowledge sits with experts, and expert attention is the least scalable resource in the pipeline: expensive, slow, and — in a marking period, a design review, a clinical workflow — strictly rationed. The interpretive step is thus simultaneously where automation is weakest and where the input needed to compensate is hardest to obtain.

Both extreme responses fail. Routing every deviation to a human removes the benefit of automation; routing none leaves the assessment resting on whatever the model inferred. What is missing is the space between: a principled account of *when* a system should spend a unit of expert attention, *what* should be retained from the answer, and *where* that answer may legitimately be applied again. In current practice the ruling is consumed once and discarded, and the next case raising the same question restarts the argument from nothing.

1.3 What already exists, and what it leaves open

VEGO-AI is the published foundation for this work, authored by the supervisors of this doctorate; the present candidate is not an author of that paper and claims none of its results. It already performs work this thesis need not redo: operationalizing modelling guidelines, evaluating models against them, feeding newly discovered valid alternatives back into the guidelines, and classifying recurring deviation patterns as substantial or occasional.

Its reported results are uneven in an informative way, and both sides of that unevenness matter here. Attributed to that paper: the generated language templates achieved F1 between 0.75 and 1.0 against human-curated ground truth across independent runs; average guideline-cluster alignment ranged from 0.70 to 0.88 across the four settings; and expert-rated compliance-vector scoring averaged 0.80 to 0.96. Against this, the expert-rated **uncovered-fragment audits** — judging whether an addition a student made but the guideline did not cover is acceptable — averaged only 0.55 to 0.88, with both use-case-diagram settings at 0.55; and the Spearman correlation between the Model Inspector's scores and a junior grader's marks was positive and statistically significant but weak ($\rho = 0.22$, $p = 0.007$). That paper's own reading of the weaker uncovered-audit figures is that this interpretive judgement may require human involvement, and its future-work section names human-in-the-loop oversight at three pipeline stages — guideline validation, compliance review, and variability classification — as a natural next step toward deployment in real assessment settings.

What the framework does not yet have is any governed means of accepting the human involvement it identifies as needed: no policy for deciding which of its many uncertain judgements warrant an expert's time; no representation for what the expert says once asked, beyond a corrected label; no record of the system reasoning the expert was responding to, and hence no way to separate disagreement with a conclusion from disagreement with the argument behind it; no provenance, scope, or revocation controls on a ruling once given; and consequently no basis either for reusing a ruling in a later case or for testing whether such reuse is safe. That is the space this doctorate occupies, and Chapter 3 develops the argument in full.

1.4 Research objective and question set

Objective. To produce and evaluate the design knowledge needed to connect selective expert intervention, structured judgment capture, and scope-aware reuse in agentic-AI assessment, together with an evaluation approach capable of stating which parts of that lifecycle transfer across contexts and which do not.

The question set below is the live wording refined during the 2026-08-05 supervision discussion. It is **provisional**, pending verification against the candidate's own record of that call and logged decisions D-RQ-01 and D-RQ-02; among the open items is whether the thesis says "human judgment" or "expert judgment" throughout.

- **U-RQ.** How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human–AI co-reasoning?
- **SQ1 — Selective intervention.** When and how, in variability exploration scenarios, should an agentic assessment system request human judgment so that important uncertainties are addressed without unnecessary expert burden?

- **SQ2 — Governed knowledge reuse.** How should expert judgment — including the system's core reasoning — be represented, validated, reconciled, and stored so it can be reused transparently without unsafe generalization or loss of human authority?
- **SQ3 — Evaluation and transfer.** How can expert judgment be reused and transferred across different guideline-operationalization contexts without unsafe generalization or loss of human authority, first in software/modelling and, when governance and access permit, in healthcare?

Each sub-question maps to exactly one study, and together they compose a lifecycle: when to ask, what to keep, where it may travel. Each carries an evaluation criterion inside its own wording, so a design meeting one goal by sacrificing its counterweight fails the question by definition. None presupposes that captured judgment *improves* assessment; whether it does, and at what cost in expert time, is an empirical outcome, and a negative result is informative.

1.5 Intended contributions

These are **intended** contributions stated with their present evidence state, kept in three kinds because they carry different evidential weight.

Design knowledge — the primary intended contribution, on which the thesis stands or falls: intervention criteria and attention-budget policies specifying when an agentic process should interrupt an expert; a representation for expert judgment that preserves case-grounding, scope, contestability, and attributable authority, and that captures the system reasoning the expert responded to; and criteria distinguishing domain-specific uncertainty from broadly transferable capability gaps. Present state: argued design rationale, not validated theory.

Artefacts — a selective-intervention architecture; a governed judgment lifecycle covering validation, conflict reconciliation, provenance, scope control, expiry and revocation; and a preregistered evaluation and workload protocol with a transfer taxonomy. A prototype exists in the software/modelling instantiation; what it supports today is *mechanism and readiness inspection* — that it runs non-destructively alongside the baseline pipeline, preserves provenance, and routes review — not effectiveness.

Empirical evidence — intended, and not yet held. The independent-expert-label gate stands at 0 of 24 generalization-safe candidate rows, against a threshold of at least 20 required for any quantitative statement. Until that count changes, no accuracy, agreement, generalization, or effort-reduction result is claimed anywhere in this proposal, and Chapter 5 should be read as status rather than results.

1.6 Scope and delimitations

The research is conducted **software/modelling first**. That programme is designed to answer all three sub-questions on its own; it is the protected default and contingent on nothing external. A healthcare instantiation would test the outer boundary of transfer and is explicitly conditional: six mandatory entry gates — use case, expert availability, data fit, ethics and institutional approval, approved environment, and study protocol — currently stand at 0 of 6 passed, and an internal checkpoint decides whether that track remains open. No clinical claim, no patient-level data work, and no dataset selection has occurred or may occur beforehand.

Out of scope: autonomous modification of the assessment pipeline by reused judgment, which remains advisory and gated; any accuracy, generalization, or effort-reduction claim ahead of the independent-label gate; and any treatment of captured judgment as ground truth — it is governed evidence, contestable by design. Internal dates in this proposal are project-control targets, not confirmed university deadlines.

1.7 Structure of the proposal

Chapter 2 — Literature Survey positions the relevant bodies of work and defines the two central constructs, variability in this thesis's sense and guideline operationalization; it is deliberately not yet written. **Chapter 3 — Gap and Research Questions** is complete: it argues why the gap is open rather than why any design is good, and positions VEGO-AI as motivating case rather than object of study. **Chapter 4 — Research Methodology** sets out the design-science approach, giving per research question the study, artefact, design process, and evaluation. **Chapter 5 — Preliminary Results** reports software/modelling work only, at the level of mechanism and readiness. **Chapter 6 — Plan** gives the work plan in approximately three-month, semester-aligned blocks across a three-year horizon, organised per research question.

Draft prepared out of sequence for structural review. To be rewritten once the research-question wording is signed off.